

Compact Multi-Valued Heuristic Search under Min Ordering

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Abstract

Multi-valued heuristics (MVHs) associate each state with a set of mutually non-dominated heuristic vectors, and are the most informative admissible heuristics available to multi-objective search. Existing MVH algorithms are locked to lexicographic (Lex) node ordering, because the dimensionality-reduction and t -discarding machinery they rely on requires it. Lex ordering in turn forces every search node to be *bound* to one concrete surviving heuristic vector, so computing a node’s priority requires a linear scan of its state’s heuristic set, repeated for every path reaching that state.

We show that this coupling is an artifact of the ordering function, not of MVHs. Our central observation is a decomposition identity: under the *Min* ordering function, the priority of a search node representing the *entire* set $\{g + h \mid h \in H(s)\}$ is obtained *exactly* in constant time from the ideal point of $H(s)$. The identity fails for Max, and for Lex it holds only through an element that the filtering step must first find. Consequently, under Min, node ordering can be decoupled from heuristic filtering.

Building on this we present C-MVH-MIN*, which (i) keeps one search node per physical path, (ii) computes an *exact* priority for that node’s whole implicit evaluation set in $O(1)$, (iii) replaces set-based t -discarding with the constant-time dominance checks of MOS-A*, thereby inheriting early termination, and (iv) retains the strictly stronger pruning power of the MVH through an existential orthant query over a full-dimensional spatial index of $H(s)$. We prove correctness for the bi-objective case and show that C-MVH-MIN* reproduces, node for node, the OPEN ordering of the fully materialised MVH search without ever materialising it.

1 Introduction

[Placeholder — to be written.]

2 Related Work

[Placeholder — to be written.]

3 Problem Formulation

We consider the multi-objective shortest-path (MOSP) problem on a directed graph $G = \langle S, E, c \rangle$, where S is a finite set of states, $E \subseteq S \times S$ is a set of directed edges, and $c : E \rightarrow \mathbb{R}_{\geq 0}^d$ assigns each edge a d -dimensional non-negative cost vector. A MOSP instance is $P = \langle S, E, c, s_{start}, s_{goal} \rangle$. A path $\pi = s_1, \dots, s_k$ has cost $c(\pi) = \sum_{i=1}^{k-1} c(s_i, s_{i+1})$.

For $x, y \in \mathbb{R}_{\geq 0}^d$ we say x *weakly dominates* y , written $x \preceq y$, if $x_i \leq y_i$ for all i ; and x *dominates* y , written $x \prec y$, if $x \preceq y$ and the inequality is strict in at least one component. Two vectors are *mutually non-dominated* if neither dominates the other. For a set X of cost vectors, $\text{ND}(X)$ denotes its cost-unique subset of mutually non-dominated vectors. For a set T we write $T \preceq x$ as shorthand for $\exists t \in T : t \preceq x$.

A *solution* is a path from s_{start} to s_{goal} . The objective of MOSP is to compute the Pareto-optimal solution set: all solutions whose cost vectors are not dominated by the cost vector of any other solution. Throughout this paper $d = 2$ (bi-objective search, BOS); Section 9 discusses $d > 2$.

3.1 Multi-Valued Heuristics

Rather than associating each state with a single heuristic vector, a *multi-valued heuristic* (MVH) is a mapping $H : S \rightarrow 2^{\mathbb{R}_{\geq 0}^d}$ where $H(s)$ is a finite set of mutually non-dominated heuristic vectors. An MVH is *admissible* if, for every state s and every path π from s to s_{goal} , there exists some $h \in H(s)$ such that $h \preceq c(\pi)$.

Note the quantifier order: admissibility guarantees that *some* member of $H(s)$ under-estimates a given path, not that *every* member does. An individual $h \in H(s)$ may over-estimate any particular path. This asymmetry governs every pruning argument in this paper.

Consider a search path reaching state s with accumulated cost vector g . Each $h \in H(s)$ induces an evaluation vector $f = g + h$, so the path is associated with the *evaluation set*

$$F(s, g) = \text{ND}(\{g + h \mid h \in H(s)\}). \quad (1)$$

An MVH search must reason about $F(s, g)$ without paying $|H(s)|$ per path.

3.2 Normalization and the Min Ordering Function

Objectives are typically measured on different scales, so before comparing them we normalize each to $[0, 1]$. Following Skyler et al. [2026], let $\min_i$ and $\max_i$ be a lower and an upper bound on objective i over the Pareto-optimal cost set; the normalization of a value x_i is

$$\varphi_i(x_i) = \frac{x_i - \min_i}{\max_i - \min_i}, \quad (2)$$

and we write $\bar{x} = (\varphi_1(x_1), \varphi_2(x_2))$. The only property we use is that each φ_i is a *strictly increasing function of its own coordinate alone*; Eq. (2) is one such map, and every result below holds for any coordinatewise strictly increasing normalization.

Definition 1 (Min ordering function). *For an evaluation vector f , define its primary Min value and its Min key as*

$$\mu(f) = \min(\bar{f}_1, \bar{f}_2), \quad \kappa(f) = (\mu(f), \max(\bar{f}_1, \bar{f}_2)).$$

A node with evaluation f precedes a node with evaluation f' iff $\kappa(f) <_{\text{lex}} \kappa(f')$.

That is, Min sorts each node’s normalized objective values in ascending order and compares the sorted vectors lexicographically [Skyler et al., 2026]. Min drives the search toward nodes that are extreme in at least one objective, alternating emphasis between objectives, so it discovers the boundaries of the Pareto front early. Crucially, and unlike Lex, Min admits *constant-time dominance checks* and an *early-termination* flag.

4 Background

4.1 Lazy MVH Search under Lex

L-NAMOA_{dr}^{*}-MVH [Wolff et al., 2026] combines MVHs with dimensionality reduction while requiring only an admissible MVH. Its key idea is to avoid creating a separate search node for every (g, h) combination: each physical path is represented by a single node that is *lazily bound* to one heuristic vector at a time. A node n at state s stores $g(n)$ and a currently selected $h(n) \in H(s)$, with $f(n) = g(n) + h(n)$, and OPEN is ordered lexicographically by f .

The binding is performed by CHOOSEH(s, g, T), which scans $H(s)$ in lexicographic order and returns the first vector whose truncated evaluation is not dominated by the current truncated goal frontier T :

$$h = \min_{\text{lex}} \{h \in H(s) \mid \neg \exists t \in T : t \preceq \text{Tr}(g + h)\}, \quad (3)$$

where $\text{Tr}(x) = (x_2, \dots, x_d)$ is the truncation operator underlying dimensionality reduction (DR), and $\text{Tr}(g + h) = \text{Tr}(g) + \text{Tr}(h)$. Discarding a heuristic because its evaluation is dominated by T is called *t-discarding*. If no vector survives, CHOOSEH returns $\perp$ and the path is discarded.

CHOOSEH is invoked twice per path: at generation, to bind a heuristic; and after extraction, when the node’s current evaluation has become dominated, to re-bind to the next lexicographically smallest usable vector, in which case the node is *re-inserted* into OPEN with a worse key.

Two structural consequences of Lex. First, DR requires an ordering condition (monotone growth of g_1) for the truncated local dominance test to be sufficient; when it fails, LOCALDOMCHECK falls back to a full-dimensional test against the untruncated closed set. Second — and this is the point we exploit — Eq. (3) returns a *specific element* of $H(s)$. The node’s priority is defined in terms of that element, so ordering and filtering are inseparable: computing the key *is* the scan. This costs $\Theta(|H(s)|)$ dominance tests for every path reaching s , repeated for different g ’s reaching the same state.

4.2 Ordering Functions and Constant-Time DCs

MOS-A* [Skyler et al., 2026] is a unifying framework in which the ordering function (OF) is a parameter. Besides Lex it studies Max and Min (Definition 1), and for the bi-objective case it derives *constant-time dominance checks* (CTDCs). For Min, each state s maintains two scalars $m_1(s), m_2(s)$, both initialized to ∞ ; a node is pruned if $f_1(n) \geq m_1(\cdot)$ or $f_2(n) \geq m_2(\cdot)$, tested against s_{goal} for the solution dominance check (SDC) and against $s(n)$ for the path dominance check (PDC). After surviving, $m_2(s(n))$ is updated when $f_1(n) < f_2(n)$ and $m_1(s(n))$ when $f_1(n) > f_2(n)$. Their Theorem 5 establishes that this rule prunes exactly the dominated nodes. A Boolean *Done* flag, evaluated only in the post-extraction check, additionally enables terminating before OPEN empties — a property Lex provably cannot have.

All of this theory is developed for *single-valued* heuristics (SVHs).

4.3 Geometric Indices for Dominance Checking

A dominance check is a geometric query: asking whether g is dominated is asking whether the lower orthant at g contains a stored vector. Anonymous [2026b] develops a dynamic k -d tree for exactly this query over the *path frontier* $F(v)$, the set of accumulated g -vectors stored at a vertex, answering orthant-emptiness in $O(n^{1-1/d})$ for $d = M - 1$, and proves that lexicographic (total-order) indices degrade to n tests from $M = 4$ on.

Two points matter for the present paper, and conflating them is a trap we want to close explicitly.

Observation 1 (Different indexed objects). *Anonymous [2026b] indexes the path frontier $F(v)$ and answers a point-versus-set query: is a new g dominated by a stored g ? We index the heuristic set $H(s)$ and answer a set-versus-set query: does any member of $g + H(s)$ escape the goal frontier? These are different sets, different queries, and — as the next observation records — different dimensions.*

Observation 2 (No collapse at $M = 2$). *Anonymous [2026b] notes that at $M = 2$ dimensionality reduction is decisive, leaving $d = 1$: the path frontier collapses to a single scalar, the smallest trailing component so far, and a dominance test is one comparison. This is why a geometric index is unnecessary for his problem in BOS, and why BOA* [Hernández et al., 2023] is so effective.*

The collapse is caused by DR. C-MVH-MIN drops DR entirely (Section 5), so $H(s)$ is never truncated: it remains a genuine two-dimensional staircase of mutually non-dominated trade-off points. Applying the $M = 2$ collapse to $H(s)$ would be a category error. Indeed, Anonymous [2026b] lists multi-valued heuristics among the questions his work leaves open.*

Observation 2 also explains why a k -d tree over $\text{Tr}(H(s))$ — as used by Anonymous [2026a], who accelerates CHOOSEH while preserving its selection semantics and its Lex + DR setting — degenerates to a sorted list in BOS: the truncation to $d - 1 = 1$ component, adopted for DR compatibility, destroys the geometry, not the number of objectives.

5 Compact MVH Representation under Min

We now develop the core of the paper. A search node is simply $n = (s, g)$; it carries *no* bound heuristic and implicitly represents the whole evaluation set $F(s, g)$ of Eq. (1).

5.1 The Ideal Point and the Min-Key Decomposition

Definition 2 (Ideal point). *For a state s with $H(s) = \{h^1, \dots, h^k\}$, the ideal point is the componentwise minimum*

$$h_{\text{id}}(s) = \left(\min_{h \in H(s)} h_1, \min_{h \in H(s)} h_2 \right).$$

For a node $n = (s, g)$ we write $f_{\text{id}}(n) = g + h_{\text{id}}(s)$.

$h_{\text{id}}(s)$ is computed once per state, when $H(s)$ is built, and is thereafter available in $O(1)$. In general $h_{\text{id}}(s) \notin H(s)$; it is the corner of the staircase, not a member of it. This is not an accident but the source of both the efficiency of Section 5 and the necessity of the MVH layer in Section 5.3.

Lemma 1 (Min-key decomposition). *Let s be a state with a finite non-empty $H(s)$, let $g \in \mathbb{R}_{\geq 0}^d$, and let $h_{\text{id}} = h_{\text{id}}(s)$. Then*

$$(a) \text{ (exactness on the primary value) } \min_{h \in H(s)} \mu(g + h) = \mu(g + h_{\text{id}}).$$

$$(b) \text{ (optimism on the full key) } \kappa(g + h_{\text{id}}) \leq_{\text{lex}} \kappa(g + h) \text{ for every } h \in H(s).$$

Proof. (a) Since φ_i is strictly increasing in its own coordinate and $h_{\text{id},i} = \min_h h_i$, we have for each i

$$\min_{h \in H(s)} \varphi_i(g_i + h_i) = \varphi_i(g_i + \min_h h_i) = \varphi_i(g_i + h_{\text{id},i}), \quad (4)$$

because a strictly increasing map commutes with taking a minimum over a finite set. Now, for any finite family of pairs, minimising a pointwise minimum is the same as minimising each coordinate separately:

$$\min_h \min(a(h), b(h)) = \min\left(\min_h a(h), \min_h b(h)\right).$$

Applying this with $a(h) = \varphi_1(g_1 + h_1)$ and $b(h) = \varphi_2(g_2 + h_2)$ and then substituting Eq. (4) into each argument gives

$$\min_h \mu(g + h) = \min(\varphi_1(g_1 + h_{\text{id},1}), \varphi_2(g_2 + h_{\text{id},2})) = \mu(g + h_{\text{id}}),$$

which is (a).

(b) Fix $h \in H(s)$. By (a), $\mu(g + h_{\text{id}}) \leq \mu(g + h)$, so the first components of the two keys are ordered. If the inequality is strict we are done. If they are equal, note that $h_{\text{id}} \preceq h$ componentwise, hence $\varphi_i(g_i + h_{\text{id},i}) \leq \varphi_i(g_i + h_i)$ for each i by monotonicity, so also $\max(g + h_{\text{id}}) \leq \max(g + h)$, giving the second components in the required order. $\square$

Part (a) is an *exact identity, not a relaxation*: the primary Min value of a node representing the entire set $F(s, g)$ is obtained in $O(1)$ from a per-state constant, with no scan and no choice. Part (b) records the residual approximation — only in the tie-breaking component — which we return to in Section 9.

Proposition 1 (The identity is specific to Min). *Part (a) of Lemma 1 fails for the Max ordering function, and holds for Lex only through an element of $H(s)$ rather than through h_{id} .*

Proof. *Max.* Let $\nu(f) = \max(\bar{f}_1, \bar{f}_2)$, take the identity normalization, $g = (0, 0)$ and $H(s) = \{(0, 10), (10, 0)\}$, so $h_{\text{id}} = (0, 0)$. Then $\min_h \nu(g + h) = \min(10, 10) = 10$, whereas $\nu(g + h_{\text{id}}) = 0$. The gap is 10 and can be made arbitrarily large.

Lex. Lexicographic order is invariant under translation, so $\min_{\text{lex}}\{g + h\} = g + \min_{\text{lex}} H(s)$; the minimiser is the lex-minimal *member* of $H(s)$, not h_{id} . Taking $H(s) = \{(1, 5), (2, 0)\}$ gives $\min_{\text{lex}} H(s) = (1, 5)$ while $h_{\text{id}} = (1, 0)$, so $g + h_{\text{id}}$ is not the Lex key of the set. $\square$

Proposition 1 is what forces the design. Under Max, the ideal point carries essentially no information about the set. Under Lex the set does have an $O(1)$ representative — but only in the *unfiltered* case. What CHOOSEH must actually return (Eq. (3)) is the lex-minimal *surviving* member, a predicate that depends on both g and the current frontier T ; that is a search, and a search over a g -dependent, T -dependent subset cannot be reduced to a per-state constant. Under Min it can, because Lemma 1(a) applies to the whole of $H(s)$ and needs no filtering at all. *This is the precise sense in which changing the ordering function removes the scan rather than merely accelerating it.*

5.2 Two Layers

Lemma 2 (h_{id} is an admissible SVH). *If H is an admissible MVH then h_{id} is an admissible single-valued heuristic: for every state s and every path π from s to s_{goal} , $h_{\text{id}}(s) \preceq c(\pi)$.*

Proof. Fix s and π . By admissibility of H there is some $h^* \in H(s)$ with $h^* \preceq c(\pi)$. By construction $h_{\text{id}}(s) \preceq h^*$ componentwise. Weak dominance is transitive, so $h_{\text{id}}(s) \preceq c(\pi)$. $\square$

Lemma 2 is stronger than MVH admissibility in one respect: whereas H only promises that *some* member under-estimates a given path, h_{id} under-estimates *every* path from s . This is what makes it safe to use as the sole basis for node ordering and for the constant-time checks, and it yields the two-layer architecture:

Layer	Driven by	Purpose
Ordering, PDC/SDC	$f_{\text{id}}(n)$, an SVH	CTDC-Min, <i>Done</i>
Extra pruning	full MVH $H(s)$	strictly stronger

The two layers are consistent: the quantity OPEN is sorted on is exactly the quantity the constant-time checks prune on.

5.3 Why the MVH Layer Is Not Redundant

If the ideal point already gives correctness, one may ask what the MVH still buys. The answer is that ideal-point pruning is *never stronger* than MVH pruning, and is sometimes strictly weaker.

Lemma 3 (One-sidedness). *Let T be a set of cost vectors and $n = (s, g)$.*

- (a) *If $T \preceq f_{\text{id}}(n)$ then $T \preceq (g + h)$ for every $h \in H(s)$.*
- (b) *The converse is false: there exist $H(s)$, g and T such that $T \preceq (g + h)$ for every $h \in H(s)$ while $T \not\preceq f_{\text{id}}(n)$.*

Proof. (a) Let $t \in T$ with $t \preceq g + h_{\text{id}}(s)$. For any $h \in H(s)$ we have $h_{\text{id}}(s) \preceq h$, hence $g + h_{\text{id}}(s) \preceq g + h$, and transitivity gives $t \preceq g + h$.

(b) Take $H(s) = \{(1, 8), (3, 5), (4, 4), (6, 2), (9, 1)\}$, so $h_{\text{id}}(s) = (1, 1)$, let $g = (6, 6)$ and $T = \{(8, 5), (5, 9)\}$. Then $f_{\text{id}} = (7, 7)$, and neither $(8, 5)$ nor $(5, 9)$ weakly dominates it. But

$$g + H(s) = \{(7, 14), (9, 11), (10, 10), (12, 8), (15, 7)\},$$

and $(5, 9)$ weakly dominates the first three while $(8, 5)$ weakly dominates the last two. $\square$

Lemma 3(a) makes the ideal-point test a sound $O(1)$ rejection rule; Lemma 3(b) shows it is incomplete, and that the missing cases are exactly those in which an MVH prunes and no SVH can. This is the formal content of the informal argument that MVHs are more informative than any single vector, and it is why we retain a genuine index over $H(s)$ rather than collapsing the MVH to h_{id} .

5.4 The Survival Oracle

We isolate everything the algorithm needs from $H(s)$ into a single abstract query, so that no correctness argument depends on the data structure chosen.

Definition 3 (Survival oracle). *For a state s , a cost vector g and a frontier T ,*

$$\text{SURVIVES}(s, g, T) = \left[\exists h \in H(s) : \neg(T \preceq g + h) \right].$$

This is an *existential* query — it asks only whether a survivor exists, not which one — and that is precisely what Min needs, since by Lemma 1 the priority does not depend on the identity of the survivor. Contrast Eq. (3), which must name one.

Algorithm 1 SURVIVES(s, g, T)

```
1: if  $T \preceq g + h_{\text{id}}(s)$  then return FALSE ▷ Test A: free, key already computed
2: if  $\neg(T \preceq g + h^{\min_1})$  or  $\neg(T \preceq g + h^{\min_2})$  then return TRUE ▷ Test B: staircase extremes
3:  $Q \leftarrow \{\text{root}(I(s))\}$ 
4: while  $Q \neq \emptyset$  do
5:   remove some  $N$  from  $Q$ 
6:   if  $T \preceq g + \text{lo}(N)$  then continue ▷ Lemma 4(a)
7:   if  $\neg(T \preceq g + \text{pt}(N))$  then return TRUE ▷ Lemma 4(b)
8:   add the children of  $N$  to  $Q$ 
9: end while
10: return FALSE
```

Instantiation for $d = 2$. Since $H(s)$ is mutually non-dominated it is a two-dimensional staircase. We store it in a static balanced spatial index $I(s)$ built once per state, in which every node N stores a member point $\text{pt}(N) \in H(s)$ together with the componentwise corner aggregates of its subtree,

$$\text{lo}(N) = (\min_{h \in N} h_1, \min_{h \in N} h_2), \quad \text{hi}(N) = (\max_{h \in N} h_1, \max_{h \in N} h_2),$$

both over *both* objectives, with no truncation. Note $\text{lo}(\text{root}) = h_{\text{id}}(s)$.

Lemma 4 (Subtree rules). *Let N be a node of $I(s)$ and let T be a frontier.*

(a) (Reject) *If $T \preceq g + \text{lo}(N)$ then $T \preceq g + h$ for every h in N 's subtree, so the whole subtree may be discarded.*

(b) (Accept) *If $\neg(T \preceq g + \text{pt}(N))$ then SURVIVES(s, g, T) holds, with witness $\text{pt}(N) \in H(s)$.*

Proof. (a) Every h in the subtree satisfies $\text{lo}(N) \preceq h$, hence $g + \text{lo}(N) \preceq g + h$; the claim follows by transitivity exactly as in Lemma 3(a). (b) $\text{pt}(N)$ is by construction an element of $H(s)$, so it is itself a witness for Definition 3. $\square$

Note carefully that there is no accept rule based on $\text{hi}(N)$: the upper corner is componentwise *larger* than every member and is therefore *more* likely to be dominated, so its survival certifies nothing. Only a stored member can witness survival. Algorithm 1 implements Definition 3 using Lemma 4.

Line 1 is *free*: $g + h_{\text{id}}(s)$ is $f_{\text{id}}(n)$, which the node already carries as its priority key. The ordering key thus doubles as the root pruning aggregate — the formal version of the intuition that the index yields the Min value for free. Line 2 tests the two extreme members of the staircase ($h^{\min_1}$ minimises h_1 , $h^{\min_2}$ minimises h_2), which are the members least likely to be dominated; when either survives, the query terminates in $O(1)$.

6 The C-MVH-Min* Algorithm

Algorithm 2 presents C-MVH-MIN*. It follows the MOS-A* skeleton [Skyler et al., 2026] with the Min OF, and differs from L-NAMOA*_{dr}-MVH in four places, marked in the pseudocode.

What changed, and why.

1. **No bound heuristic, no ChooseH.** A node is (s, g) ; its key is $g + h_{\text{id}}(s)$, obtained in $O(1)$ by Lemma 1. The $\Theta(|H(s)|)$ scan of Eq. (3) is gone from the ordering path.
2. **No DR, no t -discarding, no fallback.** LOCALDOMCHECK and its truncated/untruncated fallback are replaced by Algorithm 3, which is $O(1)$ and needs no ordering condition. As a consequence $H(s)$ is never truncated (Observation 2).
3. **No pop-then-reinsert.** A node's key is never tightened, so no node is ever re-inserted with a worse key. If SURVIVES fails the node is discarded outright.
4. **An existential MVH layer.** Algorithm 1 supplies the extra pruning power of Lemma 3(b) that the SVH layer cannot.

Algorithm 2 C-MVH-MIN***Require:** MOSP instance P ; admissible MVH H **Ensure:** Pareto-optimal solution set

```
1: for all  $s \in S$  do
2:    $m_1(s) \leftarrow \infty$ ;  $m_2(s) \leftarrow \infty$ ;  $Done(s) \leftarrow \text{FALSE}$ 
3:   build  $I(s)$  over  $H(s)$ ;  $h_{id}(s) \leftarrow \text{lo}(\text{root}(I(s)))$  ▷ ★ full  $d$ -dim, no Tr
4: end for
5:  $n_0 \leftarrow (s_{start}, \mathbf{0})$ ;  $f(n_0) \leftarrow h_{id}(s_{start})$  ▷ ★  $O(1)$  key, no CHOOSEH
6:  $OPEN \leftarrow \{n_0\}$ ;  $Sols \leftarrow \emptyset$ 
7: while  $OPEN \neq \emptyset$  do
8:    $n \leftarrow \arg \min_{\text{lex}} \{\kappa(f(n')) : n' \in OPEN\}$  ▷ Min OF
9:   if DOMINANCECHECK( $n$ , TRUE) then continue ▷ ★ CTDC, Alg. 3
10:  end if
11:  if  $\neg \text{SURVIVES}(s(n), g(n), Sols)$  then continue ▷ ★ MVH layer
12:  end if
13:  UPDATEVALUES( $n$ )
14:  if  $s(n) = s_{goal}$  then
15:     $Sols \leftarrow Sols \cup \{g(n)\}$ ; continue
16:  end if
17:  for all successors  $s'$  of  $s(n)$  do
18:     $g' \leftarrow g(n) + c(s(n), s')$ 
19:     $n' \leftarrow (s', g')$ ;  $f(n') \leftarrow g' + h_{id}(s')$ 
20:    if DOMINANCECHECK( $n'$ , FALSE) then continue
21:    end if
22:    if  $\neg \text{SURVIVES}(s', g', Sols)$  then continue
23:    end if
24:    insert  $n'$  into  $OPEN$ 
25:  end for
26: end while
27: return  $Sols$ 
```

7 Theoretical Analysis

7.1 Correctness

Our strategy is to reduce correctness to the SVH case and then argue that the MVH layer only removes nodes that cannot contribute.

Theorem 1 (Ordering and CTDC layer). *Consider C-MVH-MIN* with line 10 and line 21 of Algorithm 2 (the SURVIVES tests) removed. The resulting algorithm is exactly MOS-A* with the Min ordering function run on the single-valued heuristic h_{id} . Hence it returns the Pareto-optimal solution set.*

Proof. With those lines removed, a node's evaluation is $f(n) = g(n) + h_{id}(s(n))$, i.e. the standard SVH evaluation for the heuristic h_{id} ; OPEN is ordered by κ , which is the Min OF (Definition 1); and Algorithm 3 reproduces the CTDC rules of Skyler et al. [2026]. By Lemma 2, h_{id} is admissible. The claim is therefore an instance of their correctness result for MOS-A* under Min, including the validity of the *Done* flag. $\square$

Theorem 1 is where the design pays off: because Lemma 1 makes the set's key coincide with an SVH key, the entire CTDC apparatus — its correctness proof, its monotone extraction property, and early termination — transfers to the MVH setting without modification. Neither Wolff et al. [2026], who is confined to Lex, nor Skyler et al. [2026], who considers only SVHs, can take this step.

Theorem 2 (MVH layer is safe). *Adding the SURVIVES tests does not remove any Pareto-optimal solution.*

Algorithm 3 Constant-time DCs for Min, after Skyler et al. [2026]

```

1: function DOMINANCECHECK( $n, postExtraction$ )
2:    $f \leftarrow f(n); s \leftarrow s(n)$ 
3:   if  $postExtraction$  then
4:     if  $Done(s)$  then return TRUE
5:     end if
6:     if  $\mu(f) \geq \min(m_1(s_{goal}), m_2(s_{goal}))$  then
7:        $Done(\cdot) \leftarrow \text{TRUE}; \text{return TRUE}$  ▷ global
8:     end if
9:     if  $\mu(f) \geq \min(m_1(s), m_2(s))$  then
10:       $Done(s) \leftarrow \text{TRUE}; \text{return TRUE}$  ▷ local
11:    end if
12:  end if
13:  if  $f_1 \geq m_1(s_{goal})$  or  $f_2 \geq m_2(s_{goal})$  then
14:    return TRUE ▷ SDC
15:  end if
16:  if  $f_1 \geq m_1(s)$  or  $f_2 \geq m_2(s)$  then
17:    return TRUE ▷ PDC
18:  end if
19:  return FALSE
20: end function

21: function UPDATEVALUES( $n$ )
22:   if  $f_1(n) < f_2(n)$  then  $m_2(s(n)) \leftarrow f_2(n)$ 
23:   end if
24:   if  $f_1(n) > f_2(n)$  then  $m_1(s(n)) \leftarrow f_1(n)$ 
25:   end if
26: end function

```

Proof. Suppose C-MVH-MIN* discards $n = (s, g)$ at a SURVIVES test, and suppose for contradiction that some Pareto-optimal solution has the form $g + c(\pi)$ for a path π from s to s_{goal} . Since H is an admissible MVH there is $h^* \in H(s)$ with $h^* \preceq c(\pi)$, hence $g + h^* \preceq g + c(\pi)$.

Because the test returned FALSE, Definition 3 gives $T \preceq g + h^*$ for the frontier $T = Sols$ at that moment, so there is $t \in T$ with $t \preceq g + h^*$. By transitivity $t \preceq g + c(\pi)$. Every member of $Sols$ is the cost of an already-discovered solution, so $g + c(\pi)$ is weakly dominated by the cost of another solution, contradicting its Pareto optimality (recall dominance tests are weak, which yields the cost-unique set MOSP asks for). Correctness of Algorithm 1 as an implementation of Definition 3 follows from Lemma 4: rule (a) only discards subtrees all of whose members are dominated, and rule (b) only reports success on an actual member of $H(s)$. $\square$

Corollary 1 (Correctness of C-MVH-MIN*). *C-MVH-MIN* returns the Pareto-optimal solution set of P .*

Proof. Theorem 1 establishes soundness and completeness of the underlying search, and Theorem 2 shows that the additional tests prune only nodes with no Pareto-optimal completion. Soundness is immediate since every element added to $Sols$ is the cost of an actual path to s_{goal} . $\square$

7.2 Exact Simulation of the Materialised Search

The following statement makes precise the sense in which C-MVH-MIN* loses nothing by never constructing $F(s, g)$.

Theorem 3 (Exact ordering simulation). *Let A be the “materialised” algorithm that inserts one OPEN entry (s, g, h) for every $h \in H(s)$, ordered by $\kappa(g + h)$. For every path (s, g) , the primary Min value of the single C-MVH-MIN* node (s, g) equals the minimum primary Min value over the entire group of $|H(s)|$ entries that A creates for that path.*

Proof. Immediate from Lemma 1(a) with the node's key $f = g + h_{\text{id}}(s)$. $\square$

Corollary 2 (No Cartesian blow-up). *At any time, OPEN contains at most one entry per pair (s, g) , so $|OPEN|$ is independent of $\max_s |H(s)|$. Algorithm A may hold $|H(s)|$ entries for the same pair, and L-NAMOA_{dr}^{*}-MVH, while also holding one, may re-insert the same node up to $|H(s)|$ times over the course of the search.*

Proof. A node is created only at line 19 of Algorithm 2, once per generated path, and is never re-inserted, since no step modifies $f(n)$ after insertion. $\square$

7.3 Complexity per Path

Proposition 2. *Let $k = |H(s)|$. Per path reaching s , C-MVH-MIN^{*} performs $O(1)$ work to compute the priority key and $O(1)$ dominance tests in Algorithm 3. The SURVIVES query performs one free test (line 1) and terminates in $O(1)$ whenever line 1 or line 2 is conclusive; otherwise it visits only subtrees whose lower corner survives T , which is $O(\log k)$ when the surviving region meets a single root-to-leaf corridor, and $O(k)$ in the worst case.*

Proof. The key is $g + h_{\text{id}}(s)$ with $h_{\text{id}}(s)$ precomputed, so it costs two additions. Algorithm 3 performs a bounded number of scalar comparisons. For SURVIVES, line 1 reuses the already-computed key; line 2 performs two tests. In the descent, Lemma 4(a) prunes any subtree whose lower corner is dominated, so the visited set is contained in the set of tree nodes whose lower corner survives; the stated bounds follow. $\square$

We state the $O(k)$ worst case explicitly rather than hiding it. The relevant comparison is that L-NAMOA_{dr}^{*}-MVH pays $\Theta(k)$ *unconditionally*, for every path reaching s , in the ordering path itself. A plain merge over the sorted staircase is the honest baseline for Algorithm 1 and should be measured against it.

8 Numerical Example

Let $d = 2$ with a common scale for both objectives, so that $\varphi_1 = \varphi_2$ and, since the normalization is then a single increasing map applied to both coordinates, comparisons may be read off the raw values. Consider a state s with

$$H(s) = \{(1, 8), (3, 5), (4, 4), (6, 2), (9, 1)\},$$

a mutually non-dominated staircase, so $h_{\text{id}}(s) = (1, 1)$. Building $I(s)$ over the h_1 -sorted order gives a root storing $\text{pt} = (4, 4)$ with $\text{lo} = (1, 1)$, a left subtree $\{(1, 8), (3, 5)\}$ with $\text{lo} = (1, 5)$, and a right subtree $\{(6, 2), (9, 1)\}$ with $\text{lo} = (6, 1)$.

(i) Constant-time key, and the tie-break gap. Let $g = (5, 3)$. Then $f_{\text{id}} = (6, 4)$ and the primary Min value is $\mu(f_{\text{id}}) = \min(6, 4) = 4$. Materialising the set gives

$$g + H(s) = \{(6, 11), (8, 8), (9, 7), (11, 5), (14, 4)\}$$

with primary values 6, 8, 7, 5, 4; the minimum is 4, confirming Lemma 1(a). C-MVH-MIN^{*} obtains this from a per-state constant, without inspecting the five vectors. Note also $\kappa(f_{\text{id}}) = (4, 6)$ while the achieving materialised entry has $\kappa = (4, 14)$: the primary value is exact and only the tie-break component is optimistic, as Lemma 1(b) predicts.

(ii) Rejection at zero cost. Let $g' = (9, 8)$ and $T = \{(7, 12), (10, 6), (13, 3)\}$. Then $f_{\text{id}} = (10, 9)$ and $(10, 6) \preceq (10, 9)$, so line 1 of Algorithm 1 discards the node using only the key it already holds. This is sound and tight: all five vectors $\{(10, 16), (12, 13), (13, 12), (15, 10), (18, 9)\}$ are indeed weakly dominated by $(10, 6)$.

(iii) Acceptance in one test. Let $g = (5, 3)$ with the same T . Now $f_{\text{id}} = (6, 4)$ is not dominated, so line 1 is inconclusive. The root's stored member gives $g + \text{pt} = (9, 7)$, which no element of T weakly dominates, so line 7 returns TRUE with witness $(4, 4)$ after a single test.

(iv) **Where the MVH beats the ideal point.** Let $g'' = (6, 6)$ and $T = \{(8, 5), (5, 9)\}$. Here $f_{\text{id}} = (7, 7)$ survives, so no single-valued reasoning based on h_{id} can prune the node. The descent proceeds: the root member gives $(10, 10)$, dominated by $(5, 9)$; the left subtree gives $g'' + \text{lo} = (7, 11)$, dominated by $(5, 9)$, so both of its members are discarded by one test; the right subtree gives $g'' + \text{lo} = (12, 7)$, dominated by $(8, 5)$, discarding two more. No survivor remains, so the node is discarded — correctly, since all five evaluation vectors are dominated (Lemma 3(b)). Three dominance tests suffice where a scan would perform five, and the node is pruned even though the ideal point could not prune it.

9 Discussion

9.1 Added Value

Compared with	What C-MVH-MIN* adds
Wolff et al. [2026]	First non-lexicographic ordering for MVH search; the $\Theta(H(s))$ CHOOSEH scan is replaced by an $O(1)$ <i>exact</i> key (Lemma 1); set-based t -discarding is replaced by $O(1)$ CTDCs; DR, the ordering condition and the untruncated fallback are all eliminated; early termination becomes available; nodes are never re-inserted.
Skyler et al. [2026]	First extension of Min and its CTDCs from single-valued to multi-valued heuristics, with strictly stronger pruning (Lemma 3(b)) and one node per physical path.
Anonymous [2026a]	That work preserves the Lex + DR setting and CHOOSEH’s selection semantics, so it indexes truncated $(d - 1)$ -dimensional vectors and degenerates in BOS. Dropping DR keeps $H(s)$ fully two-dimensional, and changing the OF removes the search rather than accelerating it (Proposition 1).
Anonymous [2026b]	Complementary. He indexes the path frontier $F(v)$; we index the heuristic set $H(s)$ (Observation 1). His $M = 2$ constant-time result is a property of DR applied to $F(v)$ and does not transfer to $H(s)$ (Observation 2). He names multi-valued heuristics as an open question.

9.2 Limitations

We record the following honestly.

Weaker per-node guidance. Wolff et al. [2026] binds a specific, and therefore tighter, h to each node, whereas our key derives from the ideal point, the weakest admissible vector. Under Min this is not a choice: by Lemma 1 it is the *exact* key of the node’s whole implicit set, so any tighter key would misrepresent the set. Nevertheless, individual nodes may surface earlier than under a bound heuristic. Whether Min’s superior Pareto-front discovery rate, $O(1)$ checks and early termination outweigh this is an empirical question, and we present it as one.

Worst-case query cost. Proposition 2 admits an $O(k)$ worst case, and a plain merge over the sorted staircase is a legitimate competitor at $d = 2$ that must be measured, not assumed inferior.

Normalization. Min requires a normalization; the bounds of Eq. (2) are computed exactly for BOS by two single-objective searches and only approximated for $d > 2$ [Skyler et al., 2026].

Magnitude of early termination. Skyler et al. [2026] report that the *Done* flag reduced expansions by only a few percent. We inherit the mechanism but do not claim large gains from it.

9.3 An Open Problem: Tightened Keys

It is tempting to tighten a node’s key after extraction, re-inserting it with the minimum over the *surviving* subset of $H(s)$ rather than over all of it. This breaks the analysis. Once the live set depends on g and on the current frontier, the quantity being ordered is no longer a function of the state alone, so h_{id} ceases to be a fixed admissible SVH and Theorem 1 no longer applies; in particular the extraction-order step in the CTDC correctness argument of Skyler et al. [2026] — that a node expanded earlier has a no-larger component — fails. We therefore state the tightened-key variant as an open problem rather than an option.

9.4 Beyond Two Objectives

Two obstacles must be cleared for $d > 2$. First, constant-time dominance checks for Min beyond the bi-objective case are open in Skyler et al. [2026]; this is the primary blocker, and it concerns the ordering layer, not the MVH layer, since Lemma 1 holds verbatim in any dimension. Second, the survival oracle of Definition 3 becomes genuinely high-dimensional, and by the separation result of Anonymous [2026b] no single-coordinate total order can answer it in a sublinear number of tests once the query dimension reaches three. This is exactly where a geometric index earns its keep, and it suggests combining the two lines of work: his index over the path frontier and ours over the heuristic set, with the batched queries of $H(s)$ sharing corner annotations as he conjectures.

10 Empirical Evaluation

[Placeholder — to be written.]

11 Conclusion

We showed that the dependence of multi-valued heuristic search on lexicographic ordering is an artifact rather than a necessity. Under the Min ordering function, the priority of a search node standing for an entire evaluation set decomposes exactly through the ideal point of the heuristic set, in constant time. This decouples node ordering from heuristic filtering, which in turn allows constant-time dominance checks and early termination to be imported wholesale from the single-valued setting, while the multi-valued heuristic is retained purely as a pruning layer whose extra strength we characterised precisely. The resulting algorithm, C-MVH-MIN*, keeps one node per physical path, never re-inserts, and is correct for bi-objective search.

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